SEBASTIÁN JOSÉ CARDONA RAMÍREZ

Software Engineer | Full Stack & Backend Developer | Cloud & DevOps | Systems Architecture

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PROFESSIONAL SUMMARY

Software Engineer oriented towards backend development, system design, and software architecture, with hands-on experience engineering full stack web and mobile (PWA) applications, IoT platforms, and data pipelines. I approach development from an engineering discipline across the entire SDLC: requirement elicitation, domain and data modeling, architecture and API design, implementation, automated testing, security hardiness, and deployment. I select architectures, patterns, and technologies based on functional and non-functional requirements, prioritizing strong engineering fundamentals over transient tools.

I apply core principles including SOLID, ACID, and CAP, secure development practices aligned with OWASP, ISO, and IEEE standards, and agile methodologies such as Scrum. Experienced in implementing JWT, rotating access & refresh tokens, HttpOnly Cookies, RBAC, Redis token revocation, rate limiting, Argon2id hashing, input sanitization, XSS mitigation, and containerization with Docker to deliver secure, maintainable, high-performance, and scalable software.

TECHNICAL SKILLS & LAYERED COMPETENCIES

Languages & Core:	Python, TypeScript, JavaScript, SQL, C#
Backend & APIs:	FastAPI, Django, Express.js, Node.js, ASP.NET, REST APIs, WebSockets, SOAP/WSDL
Frontend & Mobile:	React, TypeScript, Vite, Vue.js, Progressive Web Apps (PWA), Responsive UI/UX
Architecture & Security:	Layered Architecture, Microservices, MVC, SOLID, ACID, OWASP Standards, JWT RS256/HS256, Argon2id, RBAC, AES-256
Data & Persistence:	PostgreSQL, Supabase, Redis, MySQL, SQLAlchemy, Alembic, Data Modeling, ETL & Ingestion
DevOps, Cloud & IoT:	Docker, Docker Compose, GitHub Actions, CI/CD, Git, Render, Vercel, Linux, MQTT, ESP32

FEATURED ENGINEERING PROJECTS

Car Inspection – Vehicle Inspection & Traceability PWA | *Python, FastAPI, React, TypeScript, PostgreSQL, Redis, Docker*

Special Vehicle Inspection Sector | 2026

- Designed and developed a full stack PWA to digitize industrial vehicle inspections, replacing paper logs with complete traceability, audit trails, photographic/audio evidence, maintenance histories, and automated reporting.
- Engineered a modular layered + MVC architecture separating Presentation, API, Services, Repositories, Domain, and Persistence with Python, FastAPI, PostgreSQL, SQLAlchemy, and Alembic for controlled schema migrations.
- Implemented secure auth & authorization via JWT, HttpOnly/Secure cookies, JTI, RBAC, and Argon2id, with Redis-based token rotation/blacklisting, rate limiting, input sanitization, and OWASP-aligned security headers.
- Containerized the entire multi-service stack with Docker Compose (React + TypeScript + Vite, FastAPI backend, PostgreSQL, and Redis).

LawSim Pymes – Regulatory Simulation Platform & System Architecture | *Python, Node.js, Vue.js, OWL 2, NLP, Redis, SOAP/WSDL*

Enterprise Blueprint & Regulatory Intelligence | 2025

- Architected an enterprise-grade hybrid blueprint combining Layered, Microservices, and MVC patterns to decouple standard CRUD operations from high-computation reasoning pipelines.
- Modeled regulatory domain ontologies using OWL 2 in Protégé, integrating HermiT reasoner, LEGAL-BETO (NLP), and SWRL rules to automatically infer financial & legal risk levels on Colombian SMEs.
- Specified formal microservice contracts (SOAP/WSDL), distributed caching with Redis, asymmetric JWT (RS256) auth, and AES-256 data-at-rest encryption.
- Applied UWE and OOHDM methodologies to model user-centric interaction and navigation workflows.

Industrial Remote Environmental Monitoring Platform | *Python, FastAPI, MQTT, WebSockets, PostgreSQL, Supabase, Docker*

Craft Breweries Telemetry & Monitoring | 2025 – 2026

- Designed and built a real-time IoT monitoring backend to ingest telemetry (temperature, humidity, CO_2) from ESP32 devices via MQTT, persisting data into PostgreSQL for historical analysis.
- Implemented secure REST APIs with FastAPI, JWT authentication, Argon2 hashing, RBAC, and strict schema validation for multi-role access control.
- Developed real-time WebSocket dashboards with configurable threshold alerting to detect critical risk conditions in fermentation tanks.
- Containerized development with Docker Compose and deployed to Render with managed PostgreSQL via Supabase.

EDUCATION & CERTIFICATIONS

- **Software Engineering** — Universidad Cooperativa de Colombia (UCC) (2026)
- **Climate Change: From Learning to Action** — United Nations (UN) (2025)
- **Soft Skills Strengthening Program (HCL)** — HabComLearn (2024)
- **Emotional Intelligence** — Corporación Unificada Nacional de Educación Superior (CUN) (2022)